

NileCAM81_CUOAGX

Release Notes



Version 1.9

e-con Systems

8/14/2024

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Introduction

NileCAM81 is a family of NileCAM cameras from e-con Systems with high-speed unified serial interface that carries video data, control data and power in a single coaxial cable. It uses serializer-deserializer (SerDes) technology that allows single coaxial cable interface for the camera. With coaxial cable lengths of up to 15m, NileCAM81 offers greater flexibility for the system architects to place the cameras in mechanically challenging designs of new generation embedded, automotive and autonomous driving applications. The flexibility of coaxial cable and the ability to carry video data, control data and power in a single cable makes NileCAM81 fast (low latency), flexible and yet reliable.

NileCAM81 uses Maxim's Gigabit Multimedia Serial Link (GMSL2) interface technology for the serial interface to the camera. This serial interface uses coaxial cable that carries high-speed video data from the camera, bidirectional control data between the camera and host controller, and power supply for the camera. The power to the camera is supplied from the host processor through this coaxial cable. NileCAM81 embeds the Maxim's serializer which serializes the high-speed image data and transmits over the coaxial cable. The serializer also supports the bidirectional control data communication between the camera and host controller. The on-board electronics of NileCAM81 recovers the power supplied from the host controller without contaminating the data transfers.

NileCAM81 is a 8 MP GMSL camera contains a 1/1.7" AR0821 CMOS image sensor from ON Semiconductor® along with on-board ISP. The coaxial cable is connected to Serializer Board through a rugged FAKRA connector. NileCAM81 can be interfaced with MAX9296A, MAX9296B and MAX96712 GMSL2 deserializers IC from Maxim's and also backward compatible with GMSL1 chips

Release Details

This section describes about identifying the deliverable.

- Release Version: **R03_RC1**
- Release Date: **14-AUG-2024**
- Product: **e-CAM_YUV_GMSL_PRODUCTS**

Description	Jetson AGX Orin™
Host machine version required for flashing software onto Jetson™ Development kit.	Ubuntu 22.04/20.04 (64-bit)
Sample rootfs Ubuntu operating system to run on Jetson™ Development Kit.	Ubuntu 20.04 (aarch64)
JetPack Version	JetPack 6.0
L4T version	L4T_R36.3.0
Kernel version	5.15.136-tegra
ecam_tk1_guvcview version	1.7.2-g5d361af
e-multicam version	e-multicam.elf-gdf850b2
NileCAM81 MCU Firmware Version	11NL81GMSLXXX01110fd8ae9bXXXXXXX

Change log: Updated to L4T36.3.0

- Release Version: **R02_RC3**
- Release Date: **08-JAN-2024**
- Product: **e-CAM_YUV_GMSL_PRODUCTS**

Description	Jetson AGX Xavier™/AGX Orin™
Host machine version required for flashing software onto Jetson™ Development kit.	Ubuntu 18.04/20.04 (64-bit)
Sample rootfs Ubuntu operating system to run on Jetson™ Development Kit.	Ubuntu 18.04 (aarch64)
JetPack Version	JetPack 5.1.2
L4T version	L4T_R35.4.1
Kernel version	5.10.120-ge06519c4a
ecam_tk1_guvcview version	1.7.2-g5d361af
e-multicam version	1.0
NileCAM81 MCU Firmware Version	11NL81GMSLXXX01110fd8ae9bXXXXXXX

Change log: Updated to e-CAM_YUV_GMSL_PRODUCTS release package

- Release Version: **R04_RC1**
- Release Date: **19-OCT-2023**

- Product: **NileCAM81_CUOAGX**

Description	Jetson Xavier
Host machine version required for flashing software onto Jetson AGX Xavier	Ubuntu 20.04 (64-bit) Ubuntu 14.04 is NOT recommended on the host machine.
Sample rootfs Ubuntu operating system to run on Jetson AGX Xavier.	Ubuntu 20.04 (aarch64)
JetPack Version	JetPack 5.1.2
L4T version	L4T_R35.4.1
Kernel version	5.10.104-ga43fec18c
ecam_tk1_guvcview version	1.7.2-gb429f64
e-multicam version	e-multicam.elf-v1.0
MCU Firmware Version	11NL81GMSLXXX01111dde7e65XXXXXXX

Change log: Updated to L4T35.4.1

- Release Version: **R03_RC2**
- Release Date: **07-JUN-2023**
- Product: **NileCAM81_CUOAGX**

Description	Jetson Xavier
Host machine version required for flashing software onto Jetson AGX Xavier	Ubuntu 20.04 (64-bit) Ubuntu 14.04 is NOT recommended on the host machine.
Sample rootfs Ubuntu operating system to run on Jetson AGX Xavier.	Ubuntu 20.04 (aarch64)
JetPack Version	JetPack 5.1.1
L4T version	L4T_R35.3.1
Kernel version	5.10.104-g16215e6bc
ecam_tk1_guvcview version	1.7.2-gb429f64
e-multicam version	e-multicam.elf-v1.0
MCU Firmware Version	11NL81GMSLXXX01110ec10884XXXXXXX

Change log: Updated Firmware Version.

- Release Version: **R03_RC1**
- Release Date: **24-MAY-2023**
- Product: **NileCAM81_CUOAGX**

Description	Jetson Xavier
Host machine version required for flashing software onto Jetson AGX Xavier	Ubuntu 20.04 (64-bit) Ubuntu 14.04 is NOT recommended on the host machine.
Sample rootfs Ubuntu operating system to run on Jetson AGX Xavier.	Ubuntu 20.04 (aarch64)
JetPack Version	JetPack 5.1.1
L4T version	L4T_R35.3.1
Kernel version	5.10.104-g16215e6bc
ecam_tk1_guvcview version	1.7.2-gb429f64
e-multicam version	e-multicam.elf-v1.0
MCU Firmware Version	11NL81GMSLXXX01110d23ad2bXXXXXXX

Change log: Updated to L4T35.3.1

- Release Version: **R02_RC1**
- Release Date: **22-MAR-2023**
- Product: **NileCAM81_CUOAGX**

Description	Jetson Xavier
Host machine version required for flashing software onto Jetson AGX Xavier	Ubuntu 20.04 (64-bit) Ubuntu 14.04 is NOT recommended on the host machine.
Sample rootfs Ubuntu operating system to run on Jetson AGX Xavier.	Ubuntu 20.04 (aarch64)
JetPack Version	JetPack 5.1
L4T version	L4T_R35.2.1
Kernel version	5.10.104-g018a5562a
ecam_tk1_guvcview version	1.7.2-gb429f64
MCU Firmware Version	11NL81GMSLXXX0111090e1c55XXXXXXX

Change log: Updated to L4T35.2.1.

- Release Version: **R01_RC4**
- Release Date: **10-FEB-2023**
- Product: **NileCAM81_CUOAGX**

Description	Jetson Xavier
Host machine version required for flashing software onto Jetson AGX Xavier	Ubuntu 20.04 (64-bit) Ubuntu 14.04 is NOT recommended on the host machine.

Sample rootfs Ubuntu operating system to run on Jetson AGX Xavier.	Ubuntu 20.04 (aarch64)
JetPack Version	JetPack 5.0.2
L4T version	L4T_R35.1.0
Kernel version	5.10.104-g0720c52d3
ecam_tk1_guvcview version	1.7.2-gb429f64
MCU Firmware Version	11NL81GMSLXXX01110c05bf19XXXXXXX

Change log: Added multi-camera Sync Support.

- Release Version: **R01_RC3**
- Release Date: **30-JAN-2023**
- Product: **NileCAM81_CUOAGX**

Description	Jetson Xavier
Host machine version required for flashing software onto Jetson AGX Xavier	Ubuntu 20.04 (64-bit) Ubuntu 14.04 is NOT recommended on the host machine.
Sample rootfs Ubuntu operating system to run on Jetson AGX Xavier.	Ubuntu 20.04 (aarch64)
JetPack Version	JetPack 5.0.2
L4T version	L4T_R35.1.0
Kernel version	5.10.104-g0720c52d3
ecam_tk1_guvcview version	1.7.2-gb429f64
MCU Firmware Version	11NL81GMSLXXX011107a57063XXXXXXX

- Release Version: **R01_RC2**
- Release Date: **10-NOV-2022**
- Product: **NileCAM81_CUOAGX**

Description	Jetson Xavier
Host machine version required for flashing software onto Jetson AGX Xavier	Ubuntu 20.04.4 (64-bit) Ubuntu 14.04 is NOT recommended on the host machine.
Sample rootfs Ubuntu operating system to run on Jetson AGX Xavier.	Ubuntu 20.04.04 (aarch64)
JetPack Version	JetPack 5.0.2
L4T version	L4T_R35.1.0
Kernel version	5.10.104-g0720c52d3
ecam_tk1_guvcview version	1.7.2-gd4daf13
MCU Firmware Version	11NL81GMSLXXX011108fdb6fbXXXXXXX

Known Issues

- When boot mode is toggled between HDR to Linear or vice versa, the preview freezes in ecam_tk1_guvcview application, preview will continue once resolution is switched.
- When the camera is streaming in manual exposure mode the fps displayed will not match the manual exposure value. The displaying fps will be approximately 3X times the actual fps value.
- NileCAM81_CUOAGX will only support multi-camera sync for HDR and Linear mode only in Auto exposure mode.

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

How to recover if black screen appears/preview freezes in ecam_tk1_guvcview camera window?

This is known issue. If this issue occurs, change the boot mode control from

While switching cam mode from HDR to Linear preview becomes freeze in e-multicam?

This is known issue. Close and reopen the application to overcome it.

Opening multiple instances of ecam_tk1_guvcview results in frame drops.

This is known issue. Opening multiple instances of ecam_tk1_guvcview will result in reduced framerate and reduced performance.

Note: Please refer the “Troubleshooting” section in the documents for the description of the limitations.

Download Release package

In this section, you can get the credentials for downloading the release package.

Download link:-

[Files - e-con File Sharing \(e-consystems.com\)](http://e-consystems.com)

Password:-

}{9_7~|8!?cvb,.@#

1. How can I get the updated package if any?

Please login to the [Developer Resources](#) website and download the latest release package

What's Next?

After understanding the Release Notes, you can refer to the following documents to understand more about NileCAM81_CUOAGX.

- *NileCAM81_CUOAGX Getting Started Manual*
- *NileCAM81_CUOAGX Release Package Manifest*
- *NileCAM81_CUOAGX Linux App User Manual*

Contact Us

If you need any support on NileCAM81_CUOAGX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	09-Nov-2022	Initial Draft for NileCAM81_CUOAGX	Camera Dev Team
1.1	23-Jan-2023	Updated Sensor format from 1/2 to 1/1.7	Camera Dev Team
1.2	30-Jan-2023	Updated to MCU Firmware and application version	Camera Dev Team
1.3	10-Feb-2023	Added Sync Support	Camera Dev Team
1.4	22-Mar-2023	Updated to L4T35.2.1	Camera Dev Team
1.5	24-May-2023	Updated to L4T35.3.1	Camera Dev Team
1.6	07-Jun-2023	Updated MCU Firmware Version	Camera Dev Team
1.7	19-Oct-2023	Updated to L4T35.4.1	Camera Dev Team
1.8	08-JAN-2024	Updated to e-CAM_YUV_GMSL_PRODUCTS release package	Camera Dev Team
1.9	14-Aug-2024	Updated to L4T36.3.0	Camera Dev Team